

Econometrics I

Lecture 1b: How Averages Behave (Asymptotics by Simulation)

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1. What does it mean for random variables to “converge”?
2. Law of Large Numbers: averages settle down
3. Central Limit Theorem: averages are approximately normal
4. Seeing both in simulation
5. Standard errors: the payoff

The style of this segment is the style of the course: we **state** the big theorems, **derive** what we can with the operator rules from Part 1, and **simulate** everything else. (Formal treatments: Hansen Ch. 6; Goff Appendix B.)

- ▶ Consider a sequence of numbers $\{X_i\}_{i=1}^n = (X_1, X_2, \dots, X_n)$.
 - Often people write $X_n = \{X_i\}_{i=1}^n$ but I hate this.
- ▶ We can ask what happens to some function $g(\{X_i\}_{i=1}^n)$ as $n \rightarrow \infty$.
 - Simple Example: $X_i = \frac{1}{2^i}$ and $g(\cdot)$ is the summation so that
$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{2^i} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \rightarrow 1.$$
- ▶ Sequences of **random variables** "approach" something too (other **random variables**)
- ▶ But what does **approaching** mean?

Let's define $S_n = g(\{X_i\}_{i=1}^n)$ as some **statistic** of the sequence of random variables $\{X_i\}_{i=1}^n$.

For example:

- ▶ Summation: $S_n = \sum_{i=1}^n X_i$; Sample Average: $S_n = \frac{1}{n} \sum_{i=1}^n X_i$;
- ▶ Largest Element: $S_n = \max X_i$
- ▶ Variance: $\mathbb{V}(X_1, X_2, \dots, X_n)$.
- ▶ Something complicated: $X_1 - 2X_2 + 3X_3 - 4X_4 \dots$

But our statistic could just be the identity function $S_n = \{X_i\}_{i=1}^n$ or transform $S_n = \{aX_i + b\}_{i=1}^n$

Two kinds of convergence (informally)

We need exactly two notions this semester:

- ▶ $S_n \xrightarrow{p} S$ (**convergence in probability**): as n grows, S_n is almost certainly *close to* S . Usually S is a **fixed number** — this is the sense in which an estimator is **consistent**.
- ▶ $X_n \xrightarrow{d} X$ (**convergence in distribution**): as n grows, the *histogram* of X_n settles into the distribution of X . Usually X is **normal** — this is where standard errors and t -statistics come from.
- ▶ Bonus fact (**continuous mapping**): smooth functions preserve convergence: if $S_n \xrightarrow{p} S$ then $g(S_n) \xrightarrow{p} g(S)$ for continuous g .

Formal definitions (and “almost sure” convergence, which we skip): Hansen Ch. 6, Goff Appendix B. In an hour you will have *watched* both kinds happen in simulation.

The Big Theorems

The Law of Large Numbers

Weak Law Of Large Numbers

If X_i for $i = 1, \dots, N$ are independent and identically distributed with finite mean μ and variance σ^2 , then as $N \rightarrow \infty$,

$$\bar{X}_N \xrightarrow{p} \mathbb{E}[X]$$

The Law of Large Numbers says the sample average converges in probability to its expectation.

- ▶ If we flip a fair coin enough times $\frac{1}{N} \sum_{i=1}^n X_i \xrightarrow{p} \frac{1}{2}$ heads.
- ▶ If we roll fair 6-sided die enough times $\frac{1}{N} \sum_{i=1}^n X_i \xrightarrow{p} 3.5$.
- ▶ If we randomly draw enough $\mathcal{N}(\mu, \sigma^2)$ variables on our computer $\frac{1}{N} \sum_{i=1}^n X_i \xrightarrow{p} \mu$.
- ▶ What if instead we asked $\frac{1}{N} \sum_{i=1}^n (X_i - \mu)^2 \xrightarrow{p} ?$.

Why the LLN works: you already did the hard part

In Part 1 we used **only the operator rules** to show:

$$\mathbb{E} [\bar{X}_n] = \mu \quad \text{and} \quad \mathbb{V} (\bar{X}_n) = \frac{\sigma^2}{n}$$

- ▶ The average is **centered in the right place**, and its **variance dies** as n grows. A random variable whose variance goes to zero has nowhere to go but its mean.
- ▶ That last step (“low variance $\Rightarrow$ probably close to the mean”) is made rigorous by **Chebyshev's inequality**: $\mathbb{P} (|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0$. (Full proof: Hansen 7.4; Goff B.2.)
- ▶ Notice what we did *not* need: the distribution of X_i . Any distribution with a finite mean and variance works — and versions exist for non-identical, dependent data too.

- ▶ As mentioned above, $\hat{\theta}$ is an **unbiased** estimator of θ if $\mathbb{E} [\hat{\theta}] = \theta$.
- ▶ $\hat{\theta}$ is a **consistent** estimator of θ if $\hat{\theta} \xrightarrow{p} \theta$ as $N \rightarrow \infty$
- ▶ While closely related, note that neither implies the other. Examples?

Mean squared error: bias² + variance

How good is an estimator? Measure its typical squared miss, and split it with the operator rules:

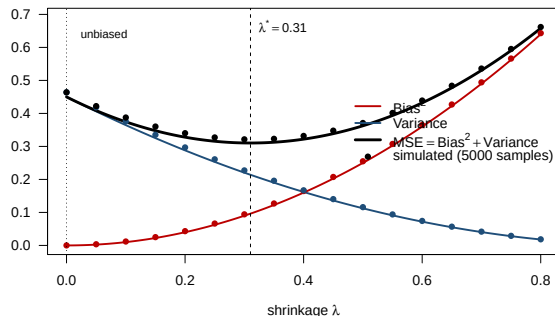
$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \underbrace{(\mathbb{E}[\hat{\theta}] - \theta)^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2]}_{\text{Variance}}$$

(Add and subtract $\mathbb{E}[\hat{\theta}]$, square; the cross term is $2 \text{Bias} \cdot \mathbb{E}[\hat{\theta} - \mathbb{E}[\hat{\theta}]] = 0$.)

- ▶ Unbiased estimators have $\text{MSE} = \text{Variance}$. That does not make them the best: a little bias that buys a lot of variance reduction is a good trade.
- ▶ Consistency in one line: if $\text{MSE}(\hat{\theta}) \rightarrow 0$ then $\hat{\theta} \xrightarrow{p} \theta$ (Chebyshev). Bias must vanish *and* variance must vanish.
- ▶ Small example: $\frac{1}{n} \sum (X_i - \bar{X})^2$ is biased for σ^2 (divide by $n - 1$ to fix it) yet has *lower* MSE than the unbiased version. Same logic on the next slide, with a knob.

The tradeoff, simulated

$X_i \sim N(\mu, \sigma^2)$, $\mu = 1$, $\sigma = 3$, $n = 20$. Shrink the sample mean toward zero: $\hat{\theta}_\lambda = (1 - \lambda)\bar{X}$, so
Bias = $-\lambda\mu$, Var = $(1 - \lambda)^2\sigma^2/n$.



- Lines: formulas. Dots: 5,000 simulated samples.
- $\lambda = 0$ is unbiased and not the minimum; shrinking by a third cuts MSE from 0.45 to 0.31.
- The catch: $\lambda^* = \frac{\sigma^2/n}{\mu^2 + \sigma^2/n}$ depends on the unknown μ . With many means at once (teachers, stores, sellers) the data supply it: empirical Bayes, Week 8.
- Ridge and lasso are this picture with λ picked by cross-validation.

- If X_i are i.i.d. draws with finite mean and variance, then as $N \rightarrow \infty$,

$$\sqrt{N} (\bar{X}_N - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

where $\mu = \mathbb{E}(X_i)$ and $\sigma^2 = \mathbb{V}(X_i)$.

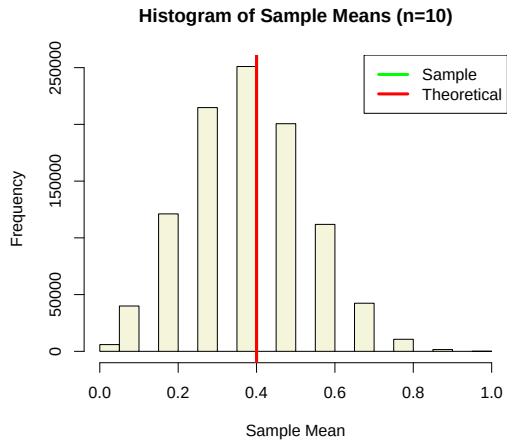
- Note that the CLT tells us something very specific about the behavior of the variance of $\bar{X}_N$ as N gets large: **root- N convergence**

- ▶ Simulate $X_i \sim \text{Bernoulli}(p)$ for $i = 1, 2, \dots, N$.
- ▶ Compute $\bar{X}_N = N^{-1} \sum_{i=1}^N X_i$. Note that
 - $\bar{X}_N$ is N^{-1} times a random variable that is $\sim \text{Binomial}(N, p)$.
 - $\mathbb{E}[\bar{X}_N] = p$.
- ▶ Repeat above $S = 1000$ times. Let $\bar{X}_{N,s}$ denote the sample mean from the s th simulation
- ▶ I'm going to do this and plot the empirical distribution (histograms) of $\bar{X}_{N,s}$ across s

```
clt_function <- function(n_sample,p){  
  # create an empty vector for the means binomial samples  
  clt <- NULL  
  n <- n_sample # sample size for each simulation  
  # take the mean of samples of the uniform distribution. repeat 1,000 times  
  for (i in 1:1000) {  
    clt <- c(clt, rbinom(1000,n_sample,p)/n_sample)  
  }  
  theoretical_mean <- p  
  hist(clt, xlim=c(0, 1), xlab='Sample Mean', main=paste0("Histogram of Sample Means (n=", eval(n_sample),")"), col='beige')  
  abline(v=mean(clt), lwd=3, col='green')  
  abline(v=theoretical_mean, lwd=3, col='red')  
  legend(c("Sample", "Theoretical"),x='topright', lwd=c(3,3), col=c('green', 'red'))  
}
```

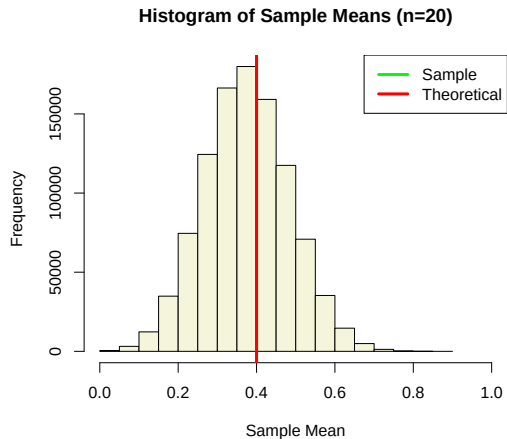
LLN Illustration

```
clt_function(10,.4)
```



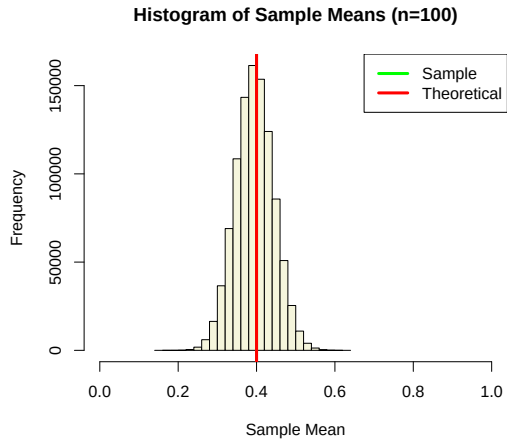
LLN Illustration

```
clt_function(20,.4)
```



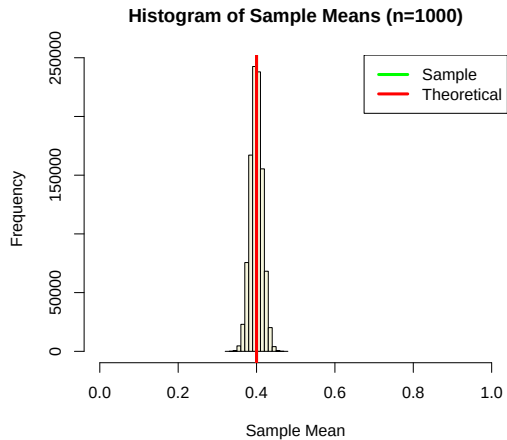
LLN Illustration

```
clt_function(100,.4)
```



LLN Illustration

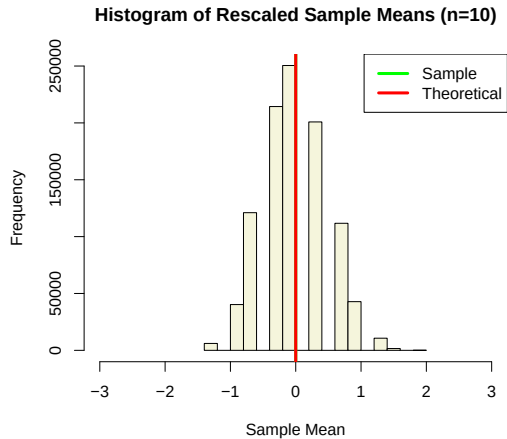
```
clt_function(1000,.4)
```



```
clt_function_scaled <- function(n_sample,p){  
  # create an empty vector for the means binomial samples  
  clt <- NULL  
  n <- n_sample # sample size for each simulation  
  # take the mean of samples of the uniform distribution. repeat 1,000 times  
  for (i in 1:1000) {  
    clt <- c(clt, (rbinom(1000,n_sample,p)/n_sample-p)*sqrt(n_sample))  
  }  
  theoretical_mean <- 0  
  hist(clt, xlim=c(-3, 3), xlab='Sample Mean', main=paste0("Histogram of Standardized Sample Means (n=", eval(n_sample),")"), col='beige')  
  abline(v=mean(clt), lwd=3, col='green')  
  abline(v=theoretical_mean, lwd=3, col='red')  
  legend(c("Sample", "Theoretical"),x='topright', lwd=c(3,3), col=c('green', 'red'))  
}
```

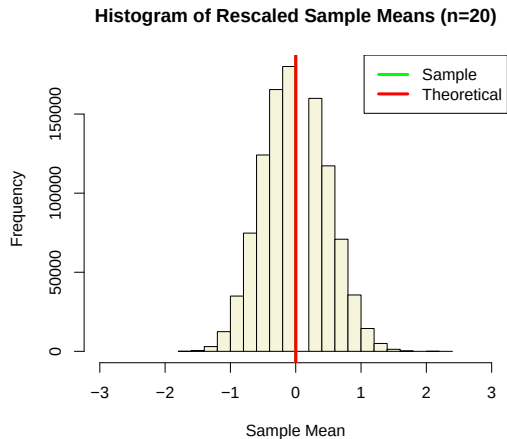
CLT Illustration

```
clt_function_scaled(10,.4)
```



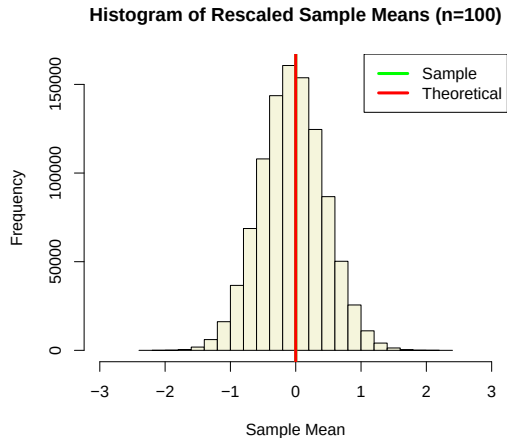
CLT Illustration

```
clt_function_scaled(20,.4)
```



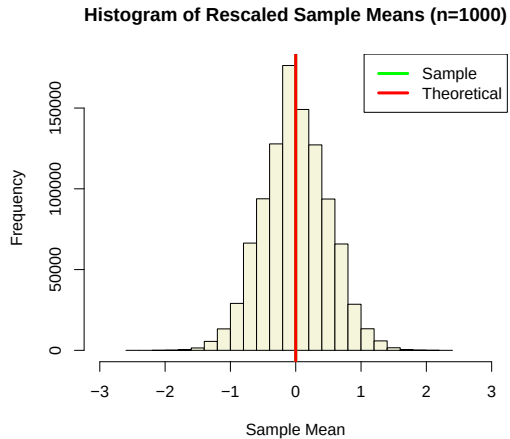
CLT Illustration

```
clt_function_scaled(100,.4)
```



CLT Illustration

```
clt_function_scaled(1000,.4)
```



$$\bar{X}_N \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{N}\right)$$

- ▶ This is **not strictly correct**: the CLT just tells us that $\bar{X}_N$ will be approximately distributed $\mathcal{N}\left(\mu, \frac{\sigma^2}{N}\right)$ as N gets large, but in any finite sample the distribution of $\bar{X}_N$
 - might not be centered around μ
 - might not have variance σ^2/N
 - might not be normal
- ▶ for i.i.d. normal case, the above statement actually is strictly correct.
- ▶ In empirical economics, we generally use this approximation, but it's important to keep in mind that it's not always a good approximation, particularly for small sample sizes when the underlying distribution of the random variable is not itself normal.
- ▶ Like the LLN, versions of the CLT exist for non-identical and dependent data (see: **ergodicity, stationarity, weak dependence**) — this matters when we get to clustered and time-series data.

- ▶ It is typically difficult to say much about the distributions of statistics in **finite samples**.
- ▶ Exception: when the underlying random variables are normally distributed. Note that above, it was trivial to derive the distribution of $\bar{X}_N$ when X_i was i.i.d. normal.
- ▶ **Monte Carlo studies** (simulation studies) are typically used to explore the finite-sample properties of estimators.
- ▶ We'll talk (briefly) later about **bootstrapped standard errors**, which can have appealing finite sample properties.

Multivariate extensions

- ▶ Let $\{\mathbf{x}_i\}$ for $i = 1, 2, \dots, \infty$ be an i.i.d. sequence of vectors of random variables with finite first and second moments.
 - Note that here we're asking for the $\mathbf{x}_i$ vector to be i.i.d. across i . Different elements of $\mathbf{x}_i$ need not be identical or independent.
- ▶ The extension of the LLN to **multivariate** random variables is trivial. We can write

$$\bar{\mathbf{x}}_N \xrightarrow{p} \mathbb{E}(\mathbf{x}_i)$$

- ▶ The multivariate CLT holds that

$$\sqrt{N}(\bar{\mathbf{x}}_N - \mathbb{E}(\mathbf{x}_i)) \xrightarrow{d} \mathcal{N}(0, \Sigma),$$

where $\Sigma = \mathbb{E}[(\mathbf{x}_i - \mathbb{E}(\mathbf{x}_i))(\mathbf{x}_i - \mathbb{E}(\mathbf{x}_i))']$

$$\sqrt{N} (\bar{X}_N - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

- ▶ We typically use the CLT to estimate the (approximate) distribution of estimators.
- ▶ This means that we need an estimate of σ^2 , and then σ^2/N estimates the variance of $\bar{X}_N$

Estimating variance

- ▶ Consider the usual setup with X_i i.i.d. for $i = 1, 2, \dots, \infty$.
- ▶ Let's show that

$$\mathbb{E} \left[(X_i - \bar{X}_N)^2 \right] = \sigma^2 \left(\frac{N-1}{N} \right)$$

- ▶ It follows that $E[s_N^2] = \sigma^2$, where

$$s_N^2 \equiv \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X}_N)^2$$

- ▶ But also notice that for large N , there is little difference between this estimator and one that divides by N .

$$s_N^2 \equiv \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X}_N)^2$$

- ▶ We can estimate $\mathbb{V}(\bar{X}_N) = \frac{\sigma^2}{N}$ with $\frac{s^2}{N}$
- ▶ A **standard error** refers to our estimate of :

$$\sqrt{\mathbb{V}(\bar{X}_N)} = \sqrt{\frac{s^2}{N}}$$

- ▶ Given the **asymptotic normality** of $\bar{X}_N$, these standard errors allow us to make statements about the (approximate) CDF of $\bar{X}_N$.
- ▶ Note that *standard errors* refer to an estimate of the *standard deviation* of an estimator.

Let's look at the workbook now.

- ▶ Open `inclass_session1.Rmd` (in the L1 - Probability folder of the course repo).
- ▶ Reproduce the simulations we just saw — run every chunk yourself.
- ▶ Then work the two problems with a neighbor:
 1. Diversification is a variance rule
 2. Total variance in a customer base

Thanks!
